TEAM ZOLLERQU

# Portfolio Selection with QAOA



Which stocks to buy? One qubit per asset, sampled on the IonQ Forte 1 noise model.

**Oussama Turki · Tushar Pandey**

This effort is supported by Qollab & IonQ

PROBLEM

# Which assets should a fund buy?

- Banks and funds build portfolios every day
- They want low risk and a good return
- Risk comes from assets that fall together

**Every extra asset doubles the choices**

| | | | |
|---|---|---|---|
| **AAPL** Apple | **GLD** gold | **XOM** oil | **JPM** bank |
| **TLT** US bonds | **JNJ** health | **PG** consumer | **XLU** utilities |
| **VNQ** real estate | **FCX** copper | **MSFT** Microsoft | **SLV** silver |

**4,096**
portfolios, 12 assets

**536,870,912**
portfolios, 29 assets

Data: weekly prices of 12 US stocks and funds, 2006-2014 (Kaggle Huge Stock Market Dataset, CC0)

FORMULATION

# One qubit per asset, one Ising energy

Score to minimise: risk minus return

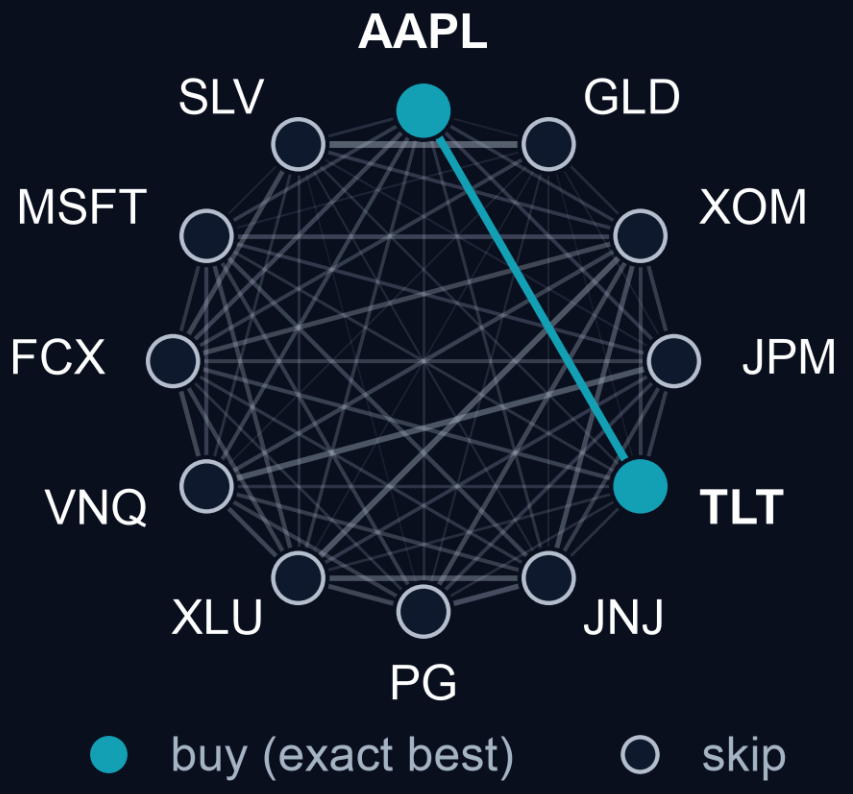
$$J(x) = \underbrace{\frac{x^{\mathrm{T}} C\, x}{n^2\, V_{\mathrm{ref}}}}_{\text{risk}} - \eta \underbrace{\frac{\mu^{\mathrm{T}} x}{n\, \sigma_{\mathrm{ref}}}}_{\text{return}}$$

The same score as an Ising energy

$$E(s) = \sum_{i<j} J_{ij}\, s_i\, s_j + \sum_i h_i\, s_i + c$$

**1 qubit = 1 asset:** |0> buy, |1> skip

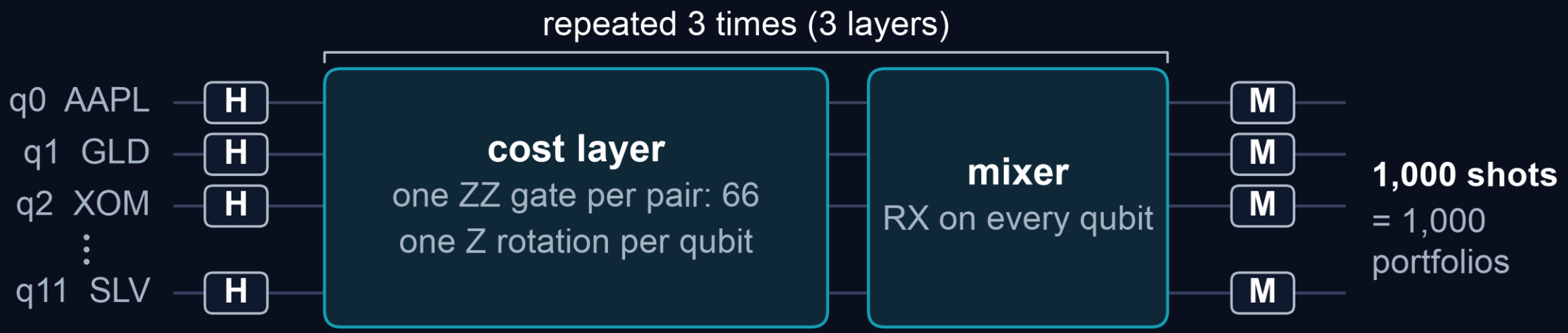
12 qubits · 66 couplings · 4,096 states




Exact best portfolio found by checking all 4,096 portfolios (classical); risk and return from 2006-2014 data

QUANTUM METHOD

# QAOA circuits, trained offline, one IonQ job




COMPARED

**8 circuit types**

58 circuits, trained offline on a classical simulator

BEST AT 12 QUBITS

**CVaR-QAOA**

3 layers, 198 two-qubit gates, sent to IonQ as one job

AT 16, 20, 29 QUBITS

**Linear-ramp QAOA**

Angles reused from 12 qubits, no new training

12 qubits: 58 circuits + a random-guess control, noiseless simulation · 16-29 qubits: server simulation

WHAT WE TRIED

# 8 circuit types, 5 settings that mattered

## 58 circuits tested

### 6 kinds of QAOA

- Linear-ramp QAOA
- Free-angle QAOA, 5 optimisers
- CVaR-QAOA (α = 0.1, 0.25)
- Multi-angle QAOA
- Counterdiabatic QAOA
- XY-mixer QAOA (keeps balance)

### 2 imaginary-time methods

- VarQITE on the QAOA angles
- varQITE ansatz (IonQ + ORNL)

## Settings we changed (noiseless)

**Mixer sign (12 assets)**
as written → opposite sign: 0.004 % → 1.8 % per shot

**Objective (12 assets)**
mean energy → CVaR 0.1: 1.8 % → 5.8 % per shot

**Layers (29 assets)**
3 → 16: 6.6 % → 90 % of runs (noiseless)

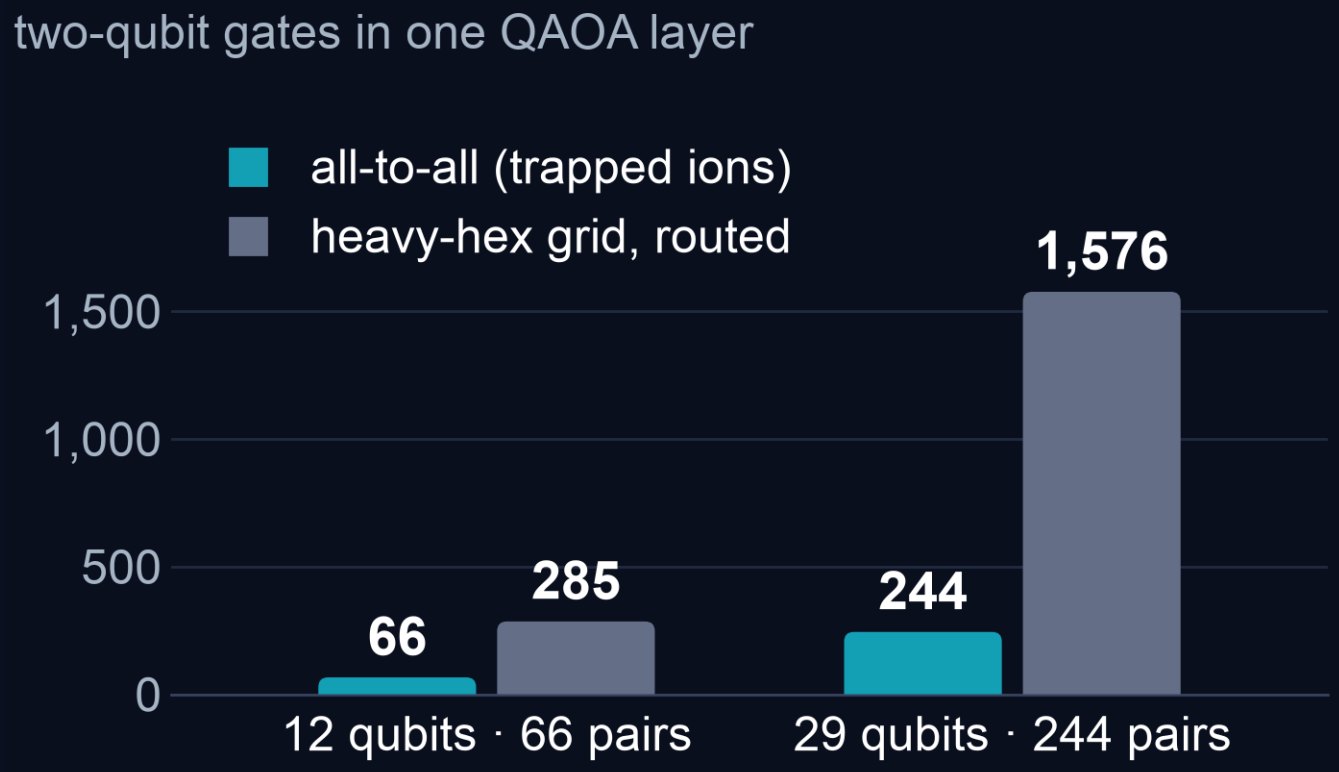
**Ramp range (29 assets)**
up to 1.5 → 3.0: 45 % → 90 % of runs at 16 layers

**Angle choice (29 assets, 3 layers)**
by energy → by CVaR: 9.9 % → 17.9 % of runs

12 assets: noiseless simulation, 10 Oct 2026 · 29 assets: noiseless server simulation, 11 Oct 2026
per shot = chance that one shot is the best portfolio · runs = 1,000-shot runs that contain it

WHY IONQ

# All-to-all links: 4.3× fewer two-qubit gates




**4.3×**

fewer two-qubit gates
at 12 qubits (66 vs 285)

**6.5×**

fewer at 29 qubits (244 vs 1,576)

66 pairs = 66 gates, no SWAPs

Gate counts from transpiling only (Qiskit 2.5.2, level 1); heavy-hex = median of 5 routing seeds on a generic heavy-hex map. No circuit was run on a heavy-hex device.

MAIN RESULT

# IonQ noise model: 29× random guessing

IonQ Forte 1 noise model

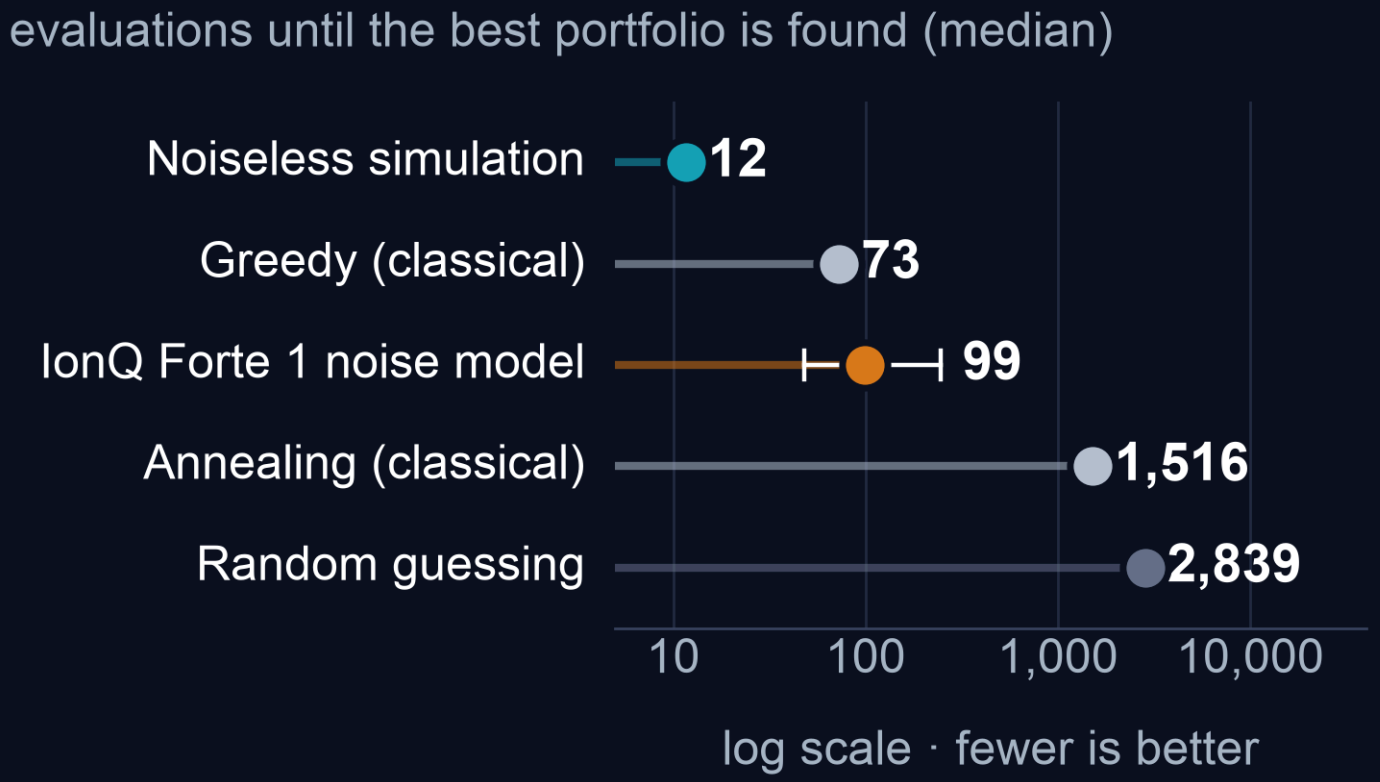
**7 / 1,000**

shots are the exact best
of 4,096 portfolios

**29×** random guessing

**58 / 1,000** noiseless

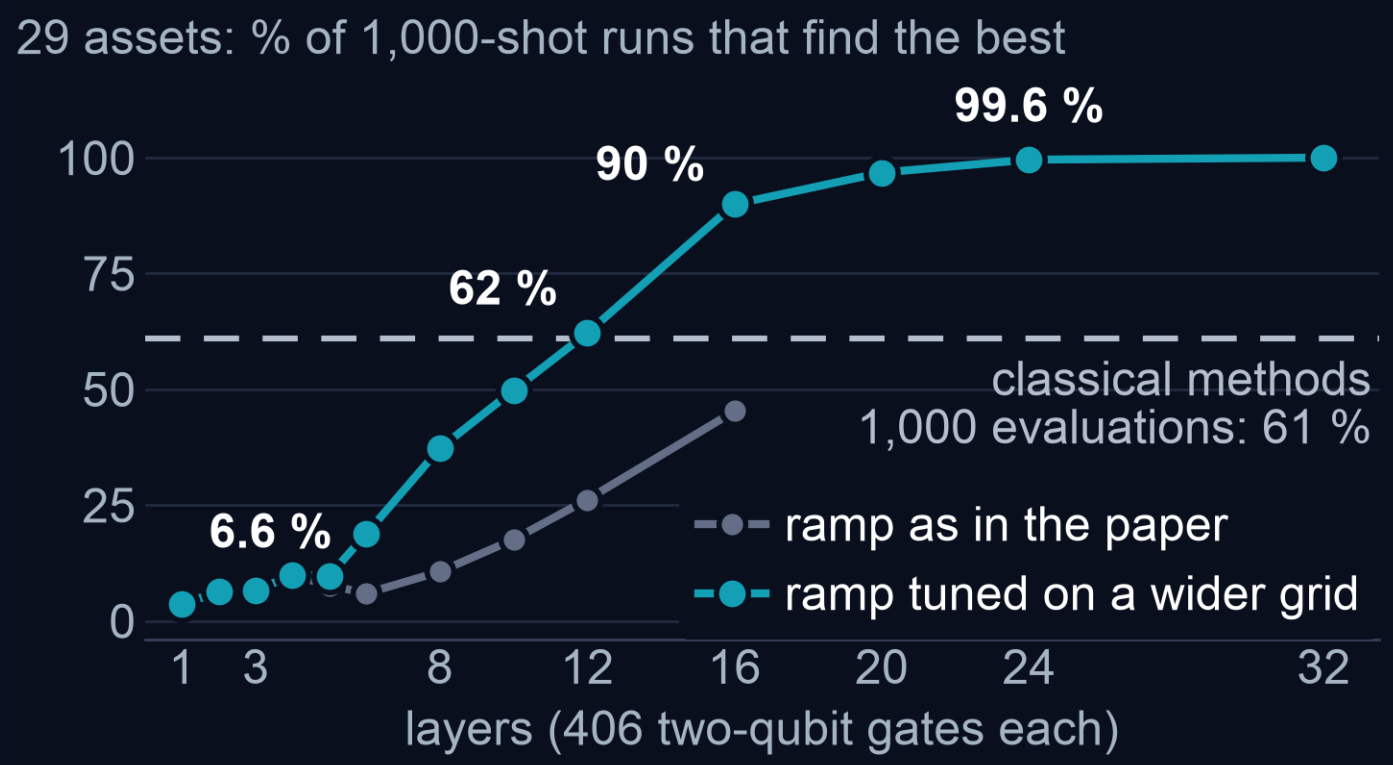
95 % interval: 3-14 per 1,000




IonQ Forte 1 noise model (Qollab): 1 job, 1,000 shots, 11 Oct 2026 · noiseless simulation
Classical: median of 50 runs. Sampler medians come from the hit rate per shot. Training not counted.

SCALING AND HONEST COMPARISON

# 29 qubits: more layers help, only without noise



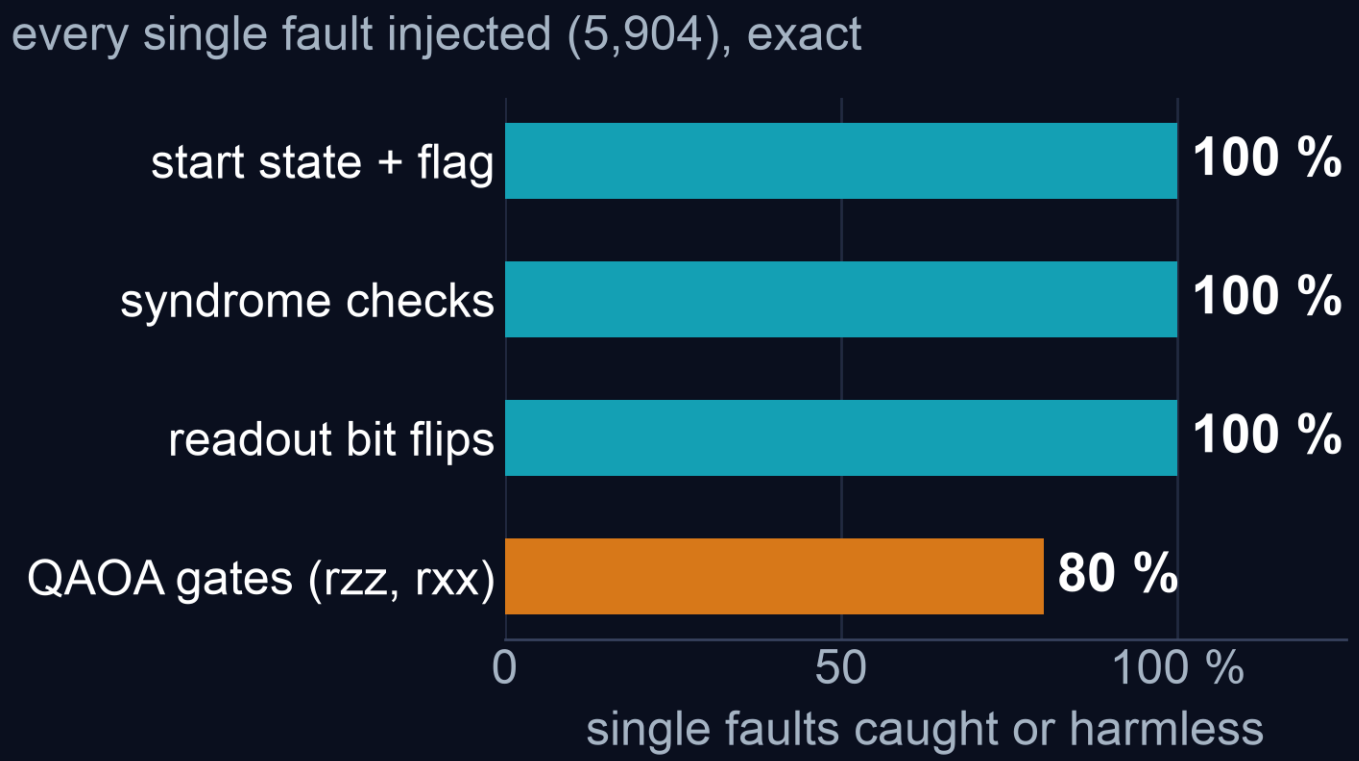

## Read with care

- Noiseless simulation only
- 16 layers = 6,496 two-qubit gates; our 198-gate IonQ run lost a factor 8 to noise
- Classical: 100 % with 10,000 evaluations
- A shot is not the same work as a classical evaluation

Server simulation of all 536,870,912 states, 11 Oct 2026; ramp tuned at 12 assets, no tuning at 29
Classical: greedy descent and annealing, 200 runs, the same 1,000 evaluations

ERROR DETECTION

# Iceberg catches errors, but doubles the gates




## 17 qubits

12 logical on 14, + 1 flag
+ 2 helpers per check

## 25 % kept

IonQ Forte 1 noise model,
1 layer: 3 in 4 shots flagged

- Best portfolio: 2 of 246 kept shots; plain 5-11 per 1,000
- 133 vs 66 two-qubit gates

Faults: exact simulation of every single fault (1, 2 layers) · Run: IonQ Forte 1 noise model, 1,000 shots
Detection, not correction: shots that fail a check are discarded (Qollab, 11 Oct 2026)

DEMO, LIMITS, NEXT STEPS

# Try it on Qollab: one click, one IonQ job

## Live demo on Qollab

1. Open the project:
   the code is in main.py
2. Run → Select QPU →
   IonQ Forte 1 (29q) noise model
3. Each job returns counts,
   a table and plots

## Honest limits

- A noisy simulator: one job of 1,000 shots
- Noise cuts the hits from 58 to 7 per 1,000
- Classical heuristics match or beat it

## Next steps

- Iceberg at more layers and checks
- Shorter circuits: 66 instead of 198 gates
- Find how many layers the noise allows

TEAM ZOLLERQU

# Thank you for your attention

**Oussama Turki · Tushar Pandey**

Portfolio Selection with QAOA



This effort is supported by Qollab & IonQ